TRASFORMATE NOTEVOLI

1) RETTANGOLO $\Pi\left(\frac{t}{\tau}\right)$

- $\tau \, \mathrm{sinc} \, f\tau$
- $\tau \, \mathrm{sinc} \, \frac{\omega\tau}{2}$

2) TRIANGOLO $\Lambda\left(\frac{t}{\tau}\right)$

- $\tau \, \mathrm{sinc}^2 \, f\tau$
- $\frac{\tau}{2} \mathrm{sinc}^2 \, \frac{\omega\tau}{4}$

3) ESPONENZIALE CAUSALE $e^{-bt}u(t)$

- $\frac{1}{b + j2\pi f}$
- $\frac{1}{b + j\omega}$

4) ESPONENZIALE SIMMETRICO $e^{-b|t|}$

- $\frac{2b}{b^2 + (2\pi f)^2}$
- $\frac{2b}{b^2 + \omega^2}$

(5) SINUSOIDE $\cos(2\pi f_c t)$

$$\frac{1}{2}\left[\delta(f-f_c)+\delta(f+f_c)\right]$$

$$\pi\left[\delta(\omega-2\pi f_c)+\delta(\omega+2\pi f_c)\right]$$

(6) IMPULSO $\delta(t-t_0)$

$$e^{-j2\pi f t_0}$$

$$e^{-j\omega t_0}$$

(7) TRENO IMPULSI $\sum_{k=-\infty}^{+\infty}\delta(t-kT_s)$

$$\frac{1}{T_s}\sum_{n=-\infty}^{+\infty}\delta\left(f-\frac{n}{T_s}\right)$$

$$\frac{2\pi}{T_s}\sum_{n=-\infty}^{+\infty}\delta\left(\omega-n\frac{2\pi}{T_s}\right)$$

(8) SEGNO $\mathrm{sgn}(t)$

$$\frac{1}{j\pi f}$$

$$\frac{2}{j\omega}$$

(9) GRADINO $u(t)$

$$\frac{1}{j2\pi f}+\frac{1}{2}\delta(f)$$

$$\frac{1}{j\omega}+\pi\delta(\omega)$$